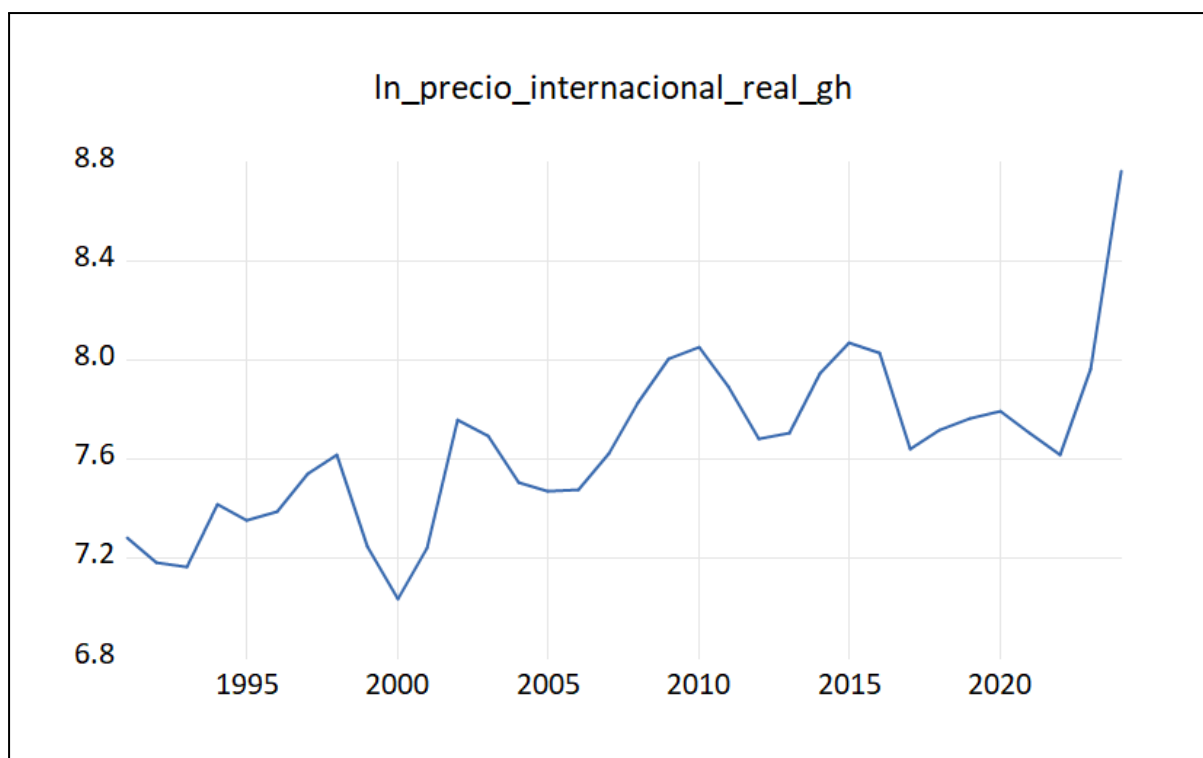




Lo primero que hago es verificar la presencia de raíces unitarias con el test de Dickey Fuller Ampliado, primero lo hago con tendencia determinística e intercepto:

Null Hypothesis: LN PRECIO INTERNACIONAL REAL GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 1 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-4.669099	0.0038
Test critical values:	1% level		-4.273277	
	5% level		-3.557759	
	10% level		-3.212361	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN PRECIO INTERNACIONAL REAL GH)				
Method: Least Squares				
Date: 09/26/26 Time: 21:04				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN PRECIO INTERNACIONAL REAL	-0.852434	0.182569	-4.669099	0.0001
D(LN PRECIO INTERNACIONAL REAL	0.790799	0.186325	4.244198	0.0002
C	6.161811	1.326327	4.645770	0.0001
@TREND("1991")	0.021405	0.005066	4.224895	0.0002
R-squared	0.500556	Mean dependent var		0.049230
Adjusted R-squared	0.447044	S.D. dependent var		0.235903
S.E. of regression	0.175420	Akaike info criterion		-0.526804
Sum squared resid	0.861616	Schwarz criterion		-0.343587
Log likelihood	12.42887	Hannan-Quinn criter.		-0.466073
F-statistic	9.354123	Durbin-Watson stat		1.543643
Prob(F-statistic)	0.000190			

Todos los coeficientes dan significativos, por lo que comparo el estadístico t con el estadístico tau al 5%, como $-4.66 < -3.55$ entonces rechazo la hipótesis nula de raíz unitaria, por lo que no es necesario diferenciar la serie.

Date: 09/26/26 Time: 20:59

Sample: 1991 2024

Included observations: 34

Included observations: 34								
Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob	
				1	0.580	0.580	12.458	0.000
				2	0.261	-0.113	15.057	0.001
				3	0.209	0.162	16.774	0.001
				4	0.303	0.200	20.524	0.000
				5	0.331	0.086	25.157	0.000
				6	0.287	0.067	28.770	0.000
				7	0.245	0.060	31.494	0.000
				8	0.281	0.125	35.215	0.000
				9	0.147	-0.203	36.277	0.000
				10	0.002	-0.113	36.277	0.000
				11	-0.053	-0.101	36.428	0.000
				12	0.020	0.001	36.450	0.000
				13	0.075	0.002	36.777	0.000
				14	0.009	-0.067	36.783	0.001
				15	-0.110	-0.073	37.563	0.001
				16	-0.174	-0.097	39.628	0.001

Como la FACP solo es significativo en el primer rezago y la FAC tarda mas en rechazarse ya que es significativo en el primer y quinto rezago entonces pruebo con un AR(1):

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/28/26 Time: 15:06

Sample (adjusted): 1992 2024

Included observations: 33 after adjustments

Convergence achieved after 3 iterations

Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.175960	0.180654	39.72215	0.0000
@TREND	0.029693	0.009205	3.225800	0.0030
AR(1)	0.527302	0.197602	2.668509	0.0122
R-squared	0.608244	Mean dependent var	7.664060	
Adjusted R-squared	0.582127	S.D. dependent var	0.336927	
S.E. of regression	0.217800	Akaike info criterion	-0.123969	
Sum squared resid	1.423108	Schwarz criterion	0.012077	
Log likelihood	5.045493	Hannan-Quinn criter.	-0.078194	
F-statistic	23.28917	Durbin-Watson stat	1.137538	
Prob(F-statistic)	0.000001			
Inverted AR Roots	.53			

Date: 09/28/26 Time: 15:07
Sample (adjusted): 1992 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*	
		1	0.262	0.262	2.4837	
		2	-0.329	-0.427	6.5123	0.011
		3	-0.318	-0.109	10.416	0.005
		4	0.001	0.010	10.416	0.015
		5	0.098	-0.084	10.814	0.029
		6	-0.004	-0.063	10.815	0.055
		7	-0.143	-0.139	11.723	0.068
		8	0.008	0.082	11.726	0.110
		9	-0.069	-0.275	11.955	0.153
		10	-0.113	-0.080	12.595	0.182
		11	-0.088	-0.149	12.999	0.224
		12	-0.011	-0.159	13.005	0.293
		13	0.081	-0.037	13.389	0.341
		14	0.047	-0.166	13.522	0.408
		15	-0.034	-0.073	13.597	0.480
		16	0.076	-0.003	13.989	0.526

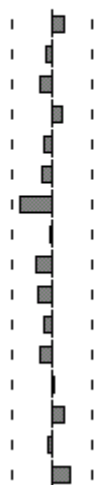
*Probabilities may not be valid for this equation specification.

Los coeficientes son significativos pero los residuos no son ruido blanco, debido a que el segundo rezago de la Función de Autocorrelaciones Parciales es significativo, por lo tanto agrego un componente AR(2) a la estimación:

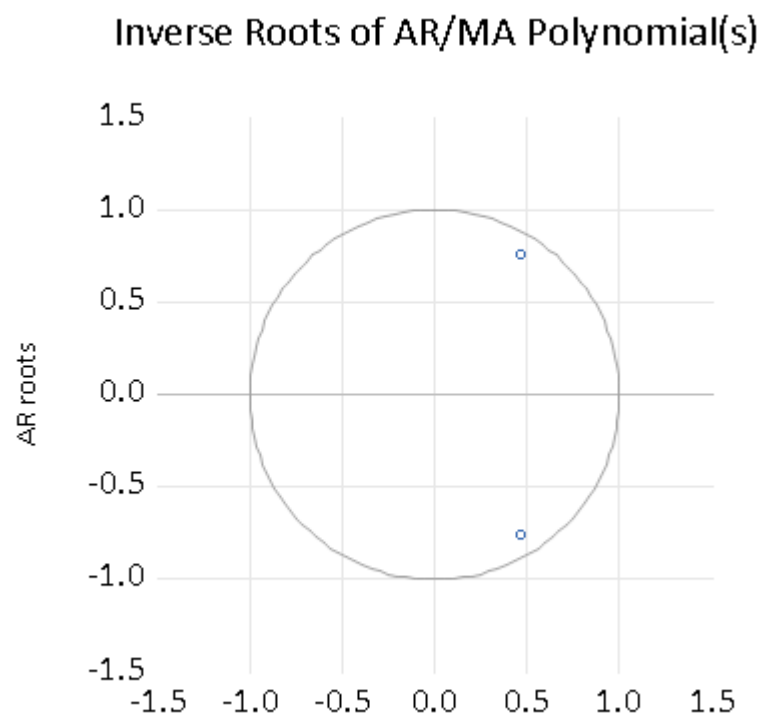
Dependent Variable: LN_PRECIO_INTERNACIONAL_REAL_GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 09/27/26 Time: 11:07
Sample (adjusted): 1993 2024
Included observations: 32 after adjustments
Convergence achieved after 2 iterations
Coefficient covariance computed using outer product of gradients

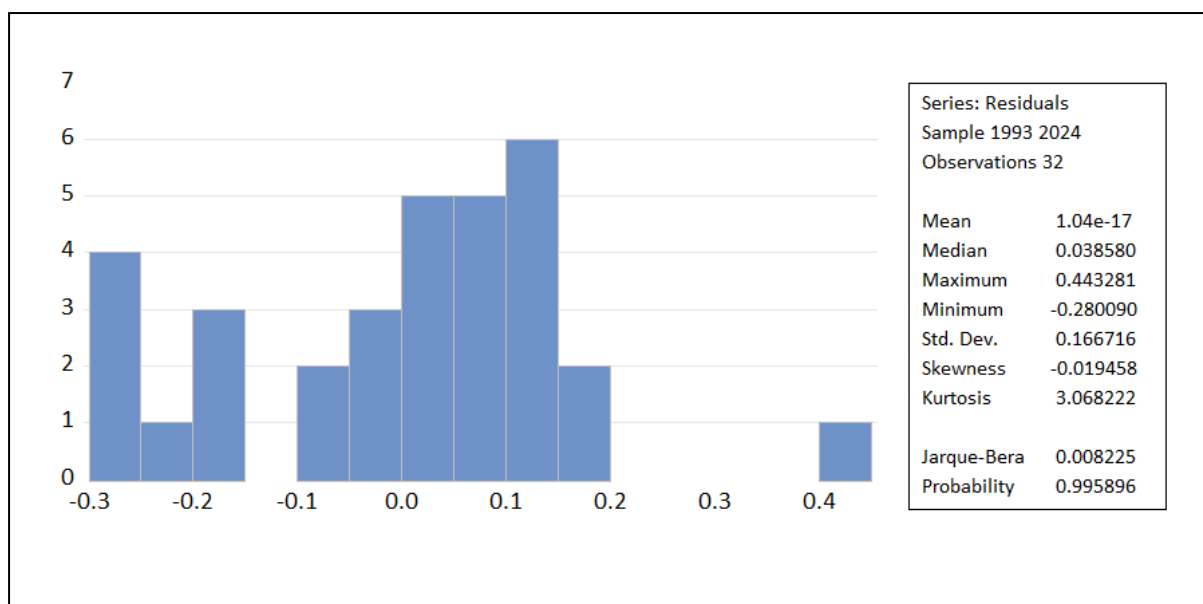
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.247435	0.075898	95.48960	0.0000
@TREND	0.025111	0.004048	6.203496	0.0000
AR(1)	0.938365	0.186313	5.036484	0.0000
AR(2)	-0.790799	0.186325	-4.244198	0.0002
R-squared	0.746185	Mean dependent var	7.679072	
Adjusted R-squared	0.718990	S.D. dependent var	0.330916	
S.E. of regression	0.175420	Akaike info criterion	-0.526804	
Sum squared resid	0.861616	Schwarz criterion	-0.343587	
Log likelihood	12.42887	Hannan-Quinn criter.	-0.466073	
F-statistic	27.43885	Durbin-Watson stat	1.543643	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.47-.76i	.47+.76i		

Date: 09/27/26 Time: 11:07
Sample (adjusted): 1993 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 0.114	0.114	0.4557	
		2 -0.052	-0.065	0.5522	
		3 -0.109	-0.096	0.9941	0.319
		4 0.090	0.113	1.3069	0.520
		5 -0.068	-0.107	1.4923	0.684
		6 -0.087	-0.070	1.8113	0.770
		7 -0.283	-0.262	5.3066	0.380
		8 -0.015	0.014	5.3163	0.504
		9 -0.143	-0.197	6.2809	0.507
		10 -0.118	-0.155	6.9637	0.541
		11 -0.076	-0.053	7.2625	0.610
		12 -0.109	-0.261	7.9143	0.637
		13 0.022	-0.022	7.9414	0.719
		14 0.113	-0.060	8.7097	0.728
		15 -0.032	-0.182	8.7744	0.790
		16 0.157	0.078	10.452	0.728

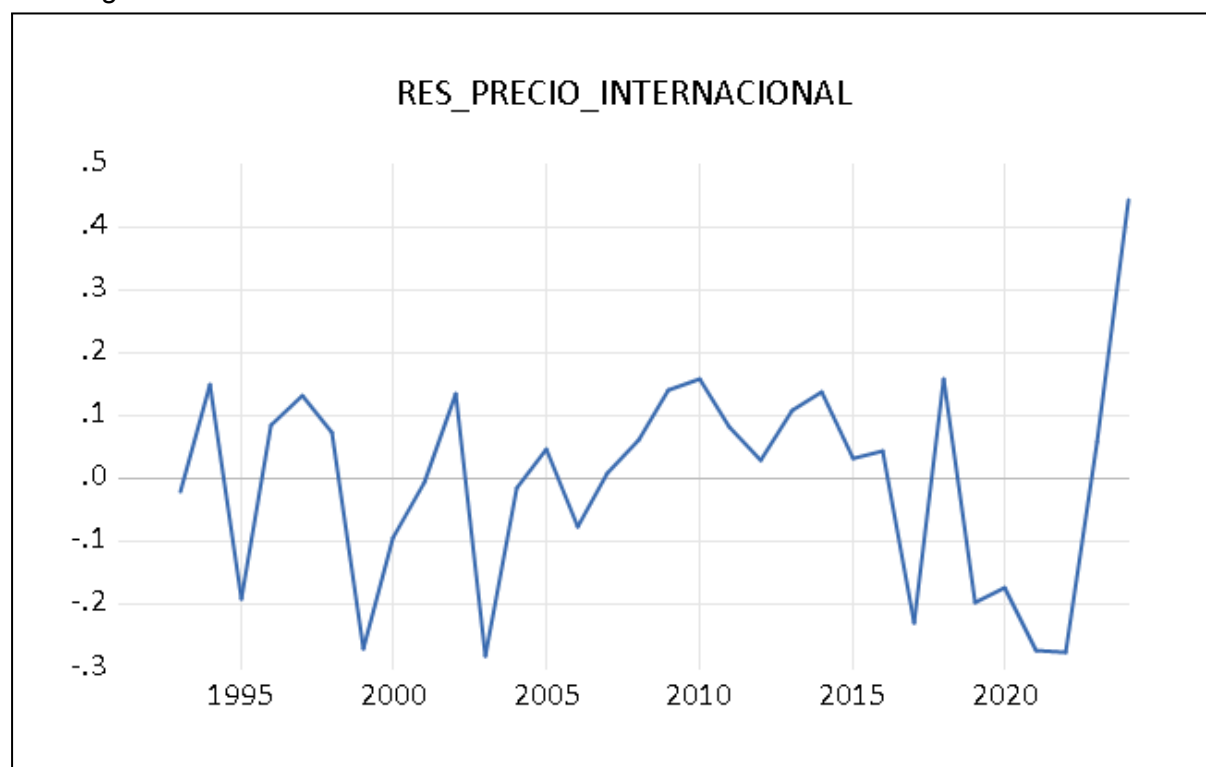
*Probabilities may not be valid for this equation specification.





Los coeficientes son significativos, el residuo es ruido blanco, las raíces están dentro del círculo unitario y los residuos se comportan de manera normal, por lo tanto el modelo a utilizar es el AR(2) con intercepto y tendencia determinística.

Para eliminar la tendencia determinística del modelo, utilizo los residuos y creo una nueva serie a partir de estos, que como son ruido blanco va a ser estacionario en torno a una media igual a 0.



Date: 10/01/26 Time: 18:01

Sample (adjusted): 1993 2024

Included observations: 32 after adjustments

Included observations: 62 after adjustments						AC	PAC	Q-Stat	Prob
Autocorrelation		Partial Correlation							
	█		█	1	0.114	0.114	0.4557	0.500	
	█		█	2	-0.052	-0.065	0.5522	0.759	
	█		█	3	-0.109	-0.096	0.9941	0.803	
	█		█	4	0.090	0.113	1.3069	0.860	
	█		█	5	-0.068	-0.107	1.4923	0.914	
	█		█	6	-0.087	-0.070	1.8113	0.936	
	█		█	7	-0.283	-0.262	5.3066	0.623	
	█		█	8	-0.015	0.014	5.3163	0.723	
	█		█	9	-0.143	-0.197	6.2809	0.712	
	█		█	10	-0.118	-0.155	6.9637	0.729	
	█		█	11	-0.076	-0.053	7.2625	0.777	
	█		█	12	-0.109	-0.261	7.9143	0.792	
	█		█	13	0.022	-0.022	7.9414	0.847	
	█		█	14	0.113	-0.060	8.7097	0.849	
	█		█	15	-0.032	-0.182	8.7744	0.889	
	█		█	16	0.157	0.078	10.452	0.842	